



Tara Chand
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B.Tech - Civil Engineering
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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology, Guwahati	8.55 (Current)	2023-Present
Senior Secondary	Central Board of Secondary Education	94.4%	2022
Secondary	Board of School Education, Haryana	92.8%	2020

EXPERIENCE

- Outsyde - Adventure Booking Platform** May 2026 – Jul 2026
Full-Stack Developer Intern Remote
 - Designed a **51-table PostgreSQL database** with **Prisma**, connecting treks, trips, bookings, payments, referrals, and payouts. Used relational constraints, price and address snapshots, soft deletes, and **append-only audit logs** to maintain data accuracy.
 - Built a **9-section admin console** with role-based permissions across **64+ APIs**, covering provider KYC, user management, refunds, disputes, commissions, and payouts. Fixed a payment race condition using **row-level locking** to prevent duplicate refunds.
 - Developed the booking platform with consumer and provider portals, **retry-safe payments** with duplicate-charge protection, and a loyalty wallet system. Added discovery features such as fuzzy search, trending rankings, and sponsored listings.

PROJECTS

- GameHub - Multi-Game Platform** | *Group Project* Mar 2026 - Apr 2026 | Website
 - Developed a multi-game platform using Node.js, Express, and MongoDB, supporting **5+ games** through unified backend architecture.
 - Secured authentication with **JWT + bcrypt** and **OTP email verification** for signup and password reset, using Brevo for emails.
 - Enabled **real-time multiplayer** for Ludo using **Socket.IO**, with turn and dice validation to prevent game-state manipulation.
 - Architected a shared game engine for reusable scoring and rules, ensuring consistent logic and adding games without backend changes.
- Smart Traffic Command Center** | *Flipkart Gridlock 2.0 Hackathon* May 2026 - Jun 2026 | Website
 - Engineered **64 key features** from **8,173** traffic incidents and trained **gradient-boosting** models, achieving **0.989 ROC-AUC**.
 - Formulated a data-driven severity score using **PCA-weighted blending** across 7 indicators, generating actionable recommendations.
 - Developed a **Streamlit** dashboard with a map predictor, model thresholds, and prediction history using crash-safe writes efficiently.
 - Established a leakage-safe pipeline using clusters, priority encodings, and training-fitted scalers, ensuring reliable model evaluation.
- TextLite - Terminal Text Editor** | *Personal Project* Jul 2026 | GitHub
 - Developed a terminal text editor in **C++17** using a singly linked list for line storage, supporting create, edit, search, and delete.
 - Structured an encapsulated **TextEditor** class, eliminating global state with explicit manual memory management using raw pointers.
 - Achieved **O(1)** line insertion and deletion through pointer-based linked-list splicing, validated with unit tests for boundary cases.
 - Integrated compile operations through a menu-driven CLI with zero external libraries, syntax highlighting, and file management.
- ForgeMind - Document Intelligence Platform** | *Personal Project* Jul 2026 | Website
 - Engineered a **knowledge retrieval engine** using FastAPI to enable **Retrieval-Augmented Generation** via knowledge extraction.
 - Integrated source-linked retrieval using MongoDB and ChromaDB to process complex unstructured text data more efficiently at scale.
 - Combined **keyword** and **semantic search** into a **hybrid document retrieval pipeline** for precise query matching and ranking.
 - Developed an **evidence-grounded answering system** with citations, temporal filtering, and checks to refuse unsupported queries.

TECHNICAL SKILLS

- Languages:** C, C++, Python, JavaScript, TypeScript
- Web & Backend:** React, Next.js, Tailwind CSS, Node.js, Express, REST APIs, WebSockets
- Databases & Services:** MySQL, PostgreSQL, MongoDB, Firebase*
- Data Science & ML:** NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, TensorFlow, Keras, LangChain*
- Core Fundamentals:** DSA, Object Oriented Programming, DBMS, OS, Probability, Statistics
- Tools:** Git, GitHub, Postman*, MATLAB*, L^AT_EX* * Elementary proficiency

ACHIEVEMENTS

- Top 75 nationally**, among **30K+** registrations, selected for the final code-a-thon at Cisco campus, Bangalore 2026
- Top 2.1%**, among **47K+** registrations, MyCareernet HackVega 2.0, national software hackathon 2026
- Semifinalist**, Flipkart GRiD 8.0, India's largest engineering challenge, Software Development Track 2026
- Semifinalist**, Flipkart Gridlock 2.0, qualified with the Smart Traffic Command Center project 2026
- Specialist** on Codeforces, Global Rank **886** among **20K+** participants in Round 1117 (Div. 2) Link
- 3★** at CodeChef, (Max. Rating **1650**) Global Rank **146** among **5K+** participants (Starters 236) Link
- Competitive Programming**, Solved **1170+** DSA problems across Codeforces, CodeChef, LeetCode & AtCoder Card

POSITION OF RESPONSIBILITY

- Core Team, Public Relations and Branding**, Alcheringa'26 (Cultural Fest), IIT Guwahati 2026
- Organiser, Technothon**, Techniche'24 (Techno-Management Fest), IIT Guwahati 2024

EXTRA-CURRICULAR ACTIVITIES

- Student Mentor**, Mentored **750+** students preparing for JEE at **Physics Wallah** 2024
- Participant**, Meta Hacker Cup, Meta's global competitive programming contest 2024
- Participant**, Mime Battle, Manthan (Inter-Hostel Cultural Competition), IIT Guwahati 2024